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Fundamentals of Data Visualization (GDV) , FS2025

1. Dataset Description

This project analyzes the **NYC 311 Service Requests** dataset (2024–2025) sourced from NYC Open Data. The dataset contains non-emergency municipal complaints, including issue types, timestamps (created and closed), and geographic identifiers such as ZIP codes and boroughs. This data is ideal for assessing city response efficiency and identifying systemic service delays at the neighborhood level.

For transparency and reproducibility, the data used in this report is available at the following links:

- **Raw Data:** NYC Open Data - 311 Service Requests. https://data.cityofnewyork.us/Social-Services/311-Service-Requests-from-2010-to-Present/erm2-nwe9/data_preview
- **Processed Dataset:** Google Drive - Cleaned NYC 311 Data. https://drive.google.com/file/d/1z434OSaGOF7Uo_rsPItQgOxD_MerdsCd/view?usp=share_link

To enable robust neighborhood-level analysis, the raw data was enriched with ZIP-to-UHF neighborhood mappings from the NYC Department of Health. This approach allows for meaningful comparisons of service responsiveness across different local communities.

2. Preprocessing Summary

The preprocessing for this project was primarily conducted during the **Data Wrangling** module. The complete cleaning pipeline and source code are documented in the following repository: https://gitlab.fhnw.ch/roberto.fazekas/DAW_NYC_data.

For the purpose of this visualization report, the following key steps and transformations were implemented:

- **Resolution Time Calculation:** Computed the duration between the creation and closure of each request. Records with logical errors (e.g., closure date appearing before the creation date) were excluded.
- **Geographic Mapping:** ZIP codes were mapped to standardized UHF (United Health Foundation) neighborhoods using NYC Department of Health references. Manual corrections were applied to ensure naming consistency across all borough definitions.
- **Outlier Management:** The **median** resolution time was used as the primary metric for aggregation to prevent extreme values from distorting the overall analysis.
- **Data Aggregation:** The dataset was grouped by neighborhood, borough, and complaint type. Rare or low-frequency complaint types were filtered out to maintain focus on statistically significant trends in the visualizations.

3. Design Rationale

With the cleaned and processed dataset—featuring standardized resolution times and robust neighborhood mappings—the next phase involves transforming this data into compelling visual narratives. The following design choices prioritize perceptual accuracy and clarity to effectively communicate key findings regarding city response efficiency and systemic service delays. The visualizations are structured to address three core analytical goals: identifying geographic disparities, isolating the most neglected issue types, and rigorously testing the effectiveness of different visual encodings.

Figure 1. NYC 311: Resolution Time by Neighborhood (Top 2 Slowest per Borough). A choropleth map illustrating the median resolution time for service requests across NYC neighborhoods. The geographic areas (UHF neighborhoods) are colored using a sequential gradient, where darker hues indicate longer resolution times. The two slowest neighborhoods in each borough are explicitly highlighted with bold outlines and text annotations containing the specific median time in hours.

Design Principles & Rationale:

- **Gestalt Principle of Proximity:** Since the data is inherently spatial, a map utilizes the viewer's natural tendency to group objects that are physically close, facilitating the identification of regional clusters of inefficiency *(Knaflic, 2020).*
- **Pre-attentive Attributes (Color):** Quantitative data is encoded using **color saturation**. Saturation is a pre-attentive attribute, allowing the brain to process magnitude (light vs. dark) prior to conscious attention *(Knaflic, 2017).*
- **Focus and Decluttering:** To avoid the "curse of expertise" (assuming the audience sees patterns instantly), the map filters information to highlight only the **Top 2 Slowest** neighborhoods. This employs the **Gestalt Principle of Enclosure** and contrast to push less critical data to the background, guiding the viewer's attention directly to the outliers (Knaflic, 2020). By reducing visual clutter—such as unnecessary gridlines or aggressive colors for non-target areas—we optimize the viewer's cognitive load *(Knaflic, 2020).*

Figure 2. Top Delayed Complaint Types. A horizontal bar chart ranking complaint types by median resolution time. The length of each bar represents the duration in months. Numerical values (e.g., "9.7 mo") are directly labeled at the end of each bar.

Design Principles & Rationale:

- **Perceptual Accuracy:** According to research on graphical perception, humans estimate **position on a common scale** (bar length) far more accurately than area or angle. Therefore, a bar chart was chosen over circular formats to precisely compare resolution times *(Bergstrom & West, 2017).*
- **Principle of Proportional Ink:** The x-axis begins strictly at zero. This adheres to the **Principle of Proportional Ink**, ensuring that the physical area of the bar is directly proportional to the numerical value it represents. Truncating the axis would distort the visual weight of the data, potentially exaggerating differences between complaint types *(Bergstrom & West, n.d.-a) (Bergstrom & West, n.d.-b).*

- **Gestalt Proximity:** Numerical values are placed directly at the end of the bars (direct labeling) rather than relying solely on a distant axis. This utilizes **Proximity** to link the label to the data point, reducing the eye movement required to read the chart and lowering cognitive effort (Knaflic, 2020).

A/B Testing: "The Biggest Pains of the City"
To validate the effectiveness of standard versus aesthetic-driven visualizations, an A/B comparison was designed using two charts representing the same data (Volume vs. Resolution Time).

Variant A: The Biggest Pains of the City (Color = Resolution Time) A horizontal bar chart using dual encoding. The length of the bars represents the Volume (number of complaints), while the color saturation of the bars encodes the Median Resolution Time (darker orange indicates longer delays).
Design Principles & Rationale:
- **Efficiency of Dual Encoding:** This design allows the viewer to simultaneously assess the frequency of a complaint and its severity. By using **length** for the primary variable (Volume), we utilize the most accurate visual channel for the most critical comparison *(Knaflic, 2020).*
- **Strategic Contrast:** Color saturation highlights specific issues like "UNSANITARY CONDITION" (dark orange) as critical problems despite having a lower volume than "Illegal Parking," facilitating a nuanced ranking task without overwhelming the viewer *(Knaflic, 2020).*

Variant B: The Biggest Pains of the City (Circle size = Resolution Time) A bubble chart (1D scatter plot) where complaint types are arranged vertically. The area of each circle (bubble) encodes the Median Resolution Time, while the position is categorical.
Design Principles (Intentional Limitations for Evaluation):
- **Inaccurate Encoding (Area):** This chart uses **circle size (area)** to encode quantitative data. This design was chosen as the "ineffective" variant for user evaluation because human perception of area is non-linear and prone to underestimation *(Bergstrom & West, n.d.-b).* It is difficult for viewers to accurately judge how much larger the "UNSANITARY CONDITION" bubble is compared to others solely by area.
- **Violation of Accuracy Principles:** While aesthetically distinct, bubble charts often prioritize form over function. Unless the area is calculated perfectly, there is a high risk of violating the **Principle of Proportional Ink** (e.g., if radius were used instead of area, the visual difference would be squared and exaggerated) *(Bergstrom & West, 2017).*

***Hypothesis*:**
We hypothesize that users will perform slower and with higher error rates when retrieving values from this chart compared to Variant A, confirming that prioritizing aesthetics over standard perceptual encodings increases cognitive load.

4. Evaluation Results

Method & Participants To validate the effectiveness of the visualizations, a **Task-Based User Evaluation** combined with an **A/B test** was conducted. The evaluation involved **5 participants** (Yaren, Ivan, Nadya, Tyoma, Angel) representing a half-expert audience. The primary metric was **time-on-task** (measured in seconds) and the accuracy of information retrieval. This method is considered reliable as it focuses on objective performance (retrieving specific information) rather than subjective preference.

Tasks & Procedure Participants were asked to perform specific data retrieval tasks using a "Think-Aloud" protocol.

To ensure consistency, **all participants were assigned the same set of questions** across all figures. To eliminate the **"order effect"** (where participants might perform better on the second chart due to familiarity with the dataset), the presentation order for the A/B testing was randomized:

• Half of the participants evaluated **Variant A** (Bar Plot) before Variant B.

• The other half started with **Variant B** (Bubble Chart) before moving to Variant A.

The specific tasks assigned were as follows:

1. **Figure 1:** Participants had to find the neighborhood in the Bronx with the longest waiting time and determine if Staten Island has neighborhoods with a waiting time of more than 50 hours.
2. **Figure 2:** The tasks involved determining the number of unjustified delay types and the resolution time for smoking complaints.
3. **A/B Testing (Variant A vs. Variant B):** Participants were offered identical questions for two types of graphs (comparing the volume of complaints and determining the top 3 fastest/slowest categories in terms of resolution time) to determine which format was more efficient.

Findings

• **Figure1 :** The map proved highly effective. Most participants completed tasks quickly (ranging from **7 to 16 seconds**) with high accuracy. The use of color intensity allowed for rapid identification of problem areas and labels allowed to name correct neighborhoods, confirming that the pre-attentive attributes of the map were functioning correctly.

• **Figure 2:** A usability issue was identified regarding color interpretation. Participants (e.g., Nadya, Tyoma) struggled to identify which colors represented "unjustified" delays, taking up to **1 minute** to complete the task. The lack of explicit labeling for the color logic caused confusion, as the meaning of bright vs. muted colors was not immediately intuitive.

• **A/B Test Results (Variant A vs. Variant B):** The comparison revealed a significant performance gap.

  ◦ **Variant A (Bar Plot):** Participants answered questions in **3–14 seconds**.

  ◦ **Variant B (Bubble/Scatter Plot):** Participants took **14–45 seconds** and made some errors.

  ◦ **Key Insight:** The difficulty with Graph B was not solely due to comparing areas. While the bubbles contained text indicating *time*, they lacked text indicating *volume* (quantity). This forced users to cognitively estimate and compare the positions of the circles without numerical support, whereas the Bar Plot allowed for instant length and values comparison, reducing the cognitive effort required.

5. Improvements from evaluation

Based on the testing results and identified interpretation difficulties, changes were introduced to the final design to improve clarity and reduce task completion time.

1. Geospatial Visualization Validation (Figure 1)
Decision: It was decided to leave **Figure 1 (Map)** unchanged.

Evidence & Rationale: During the evaluation, the map demonstrated high efficiency: users quickly and accurately located the required areas. This confirms that the chosen data encoding through color saturation and area labels worked as an effective pre-attentive attribute, allowing for instant identification of problem areas without cognitive overload. Since the design proved its reliability, no modifications were needed.

2. Adding Annotations to the Delay Graph (Figure 2 → Figure 2.1)
Change: Figure 2 was updated to version **Figure 2.1**. Top Delayed Complaint Types. Horizontal bar chart where length encodes resolution time in months. Orange indicates unjustified delays, while grey indicates normal resolution time based on the new added legend.

Why: The evaluation revealed that without hints, users were confused about the meaning of the colors, pointing to a "curse of expertise" problem — where the author assumes the encoding is obvious, but the audience does not. According to the "Show and Tell" principle, visualization becomes more effective when graphical elements are accompanied by explanatory text that guides the reader's attention and forms the correct context. The annotations eliminated ambiguity in perception.

3. **Choosing the Visualization Type based on A/B Testing Results Decision:** Based on the comparative testing results, **Variant A (Bar Plot)** was selected for the final report, effectively replacing Variant B (Bubble/Scatter Plot).

Why: The evaluation revealed that **Variant A (Bar Plot)** facilitated significantly faster data retrieval. While participants were able to identify extreme values (e.g., top 3 highest/lowest) in **Variant B** by relying on bubble size and color intensity, this format introduced unnecessary cognitive friction. The critical design flaw in Variant B was the lack of a **zero-baseline** on the axis. Participants struggled to accurately judge the position of the bubbles because the axis was truncated (did not start at 0), forcing them to perform mental calculations to understand the relative differences. In contrast, Variant A employed a standard zero-baseline, allowing users to instantly compare **length** on a common scale. This empirical finding aligns with visualization best practices, which state that bar charts must include zero to ensure that the visual weight of the graphic is proportional to the data values (Principle of Proportional Ink) and to avoid misleading comparisons

Declaration of AI Usage

- **Gemini** - Assistance in structuring the report sections, refining English grammar and academic tone, and cross-checking the design rationale against visualization principles. URL: https://gemini.google.com/
- **GitHub Copilot** - Used for assistance with the coding implementation to achieve the required technical level for the charts and data transformations. URL: https://github.com/features/copilot

Design sources

- Bergstrom, C., & West, J. (2017, April 19). *Calling Bullshit 6.1: Dataviz in the popular media* [Video]. YouTube. https://www.youtube.com/watch?v=T-5aLbNeGo0
- Bergstrom, C., & West, J. (n.d.). *Misleading axes on graphs*. Calling Bullshit. https://callingbullshit.org/tools/tools_misleading_axes.html
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- Knaflic, C. N. (2017, May 11). *How to use color in your data visualization* [Video]. YouTube. https://www.youtube.com/watch?v=AiD6etOB6qI
- Knaflic, C. N. (2020, June 4). *How to declutter data visualizations (5 steps)* [Video]. YouTube. https://www.youtube.com/watch?v=X79o46W5pII

Figure 1. NYC 311: Resolution Time by Neighborhood
Top 2 Slowest per Borough

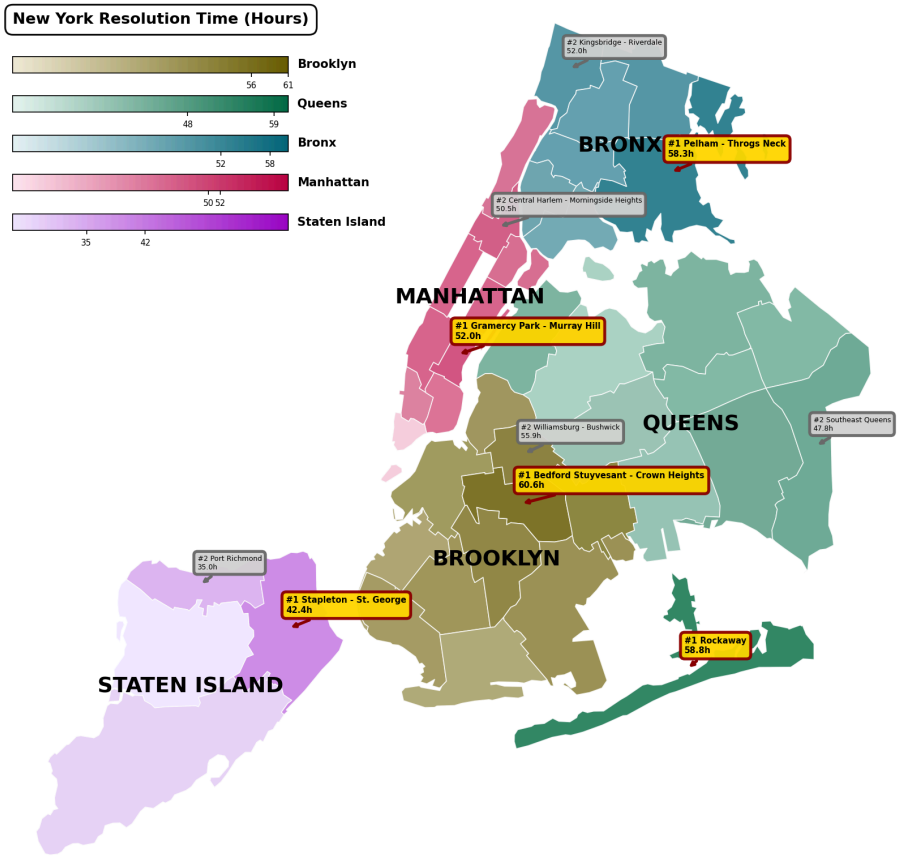


Figure 2. Top Delayed Complaint Types

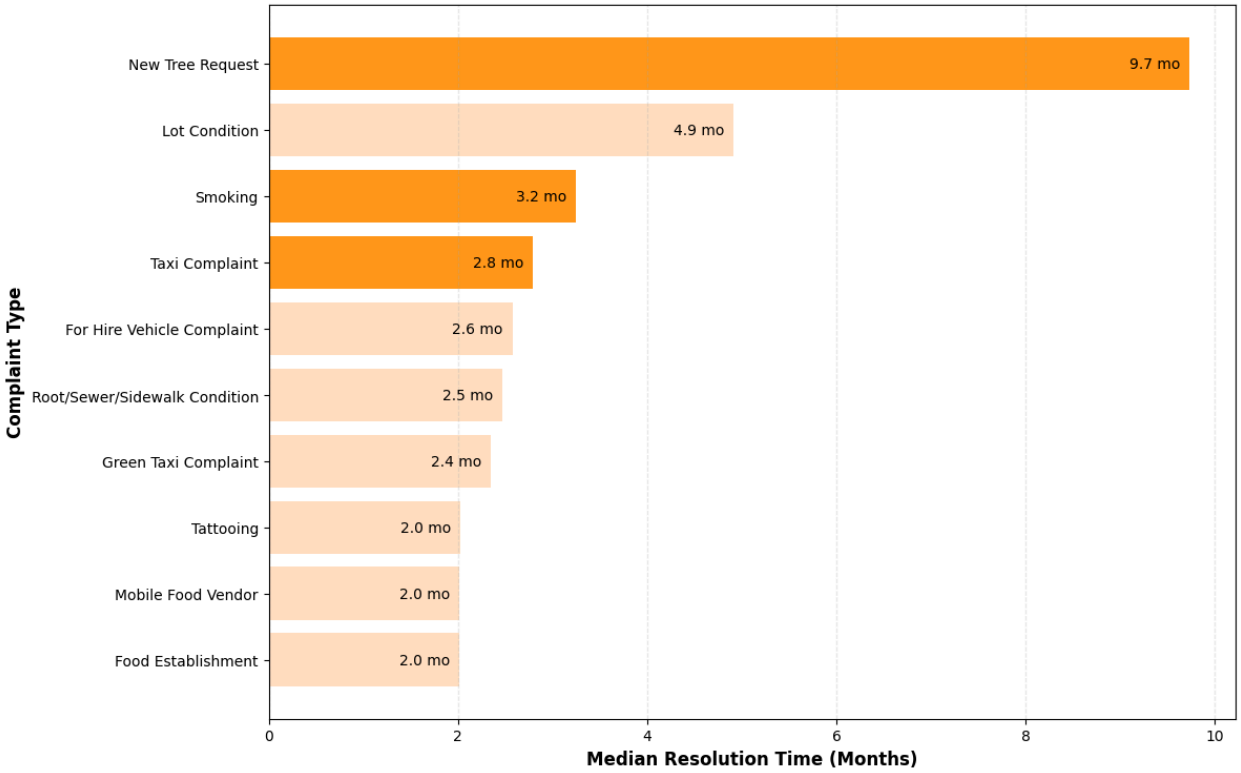
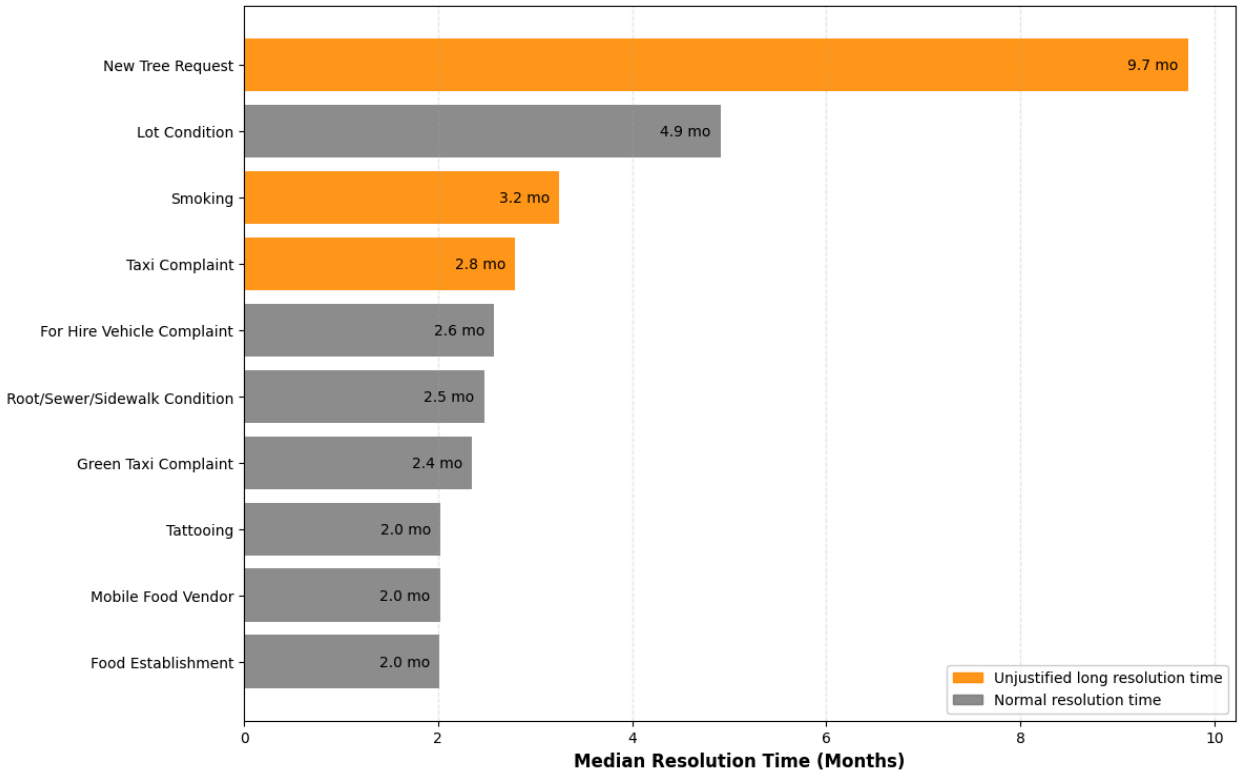
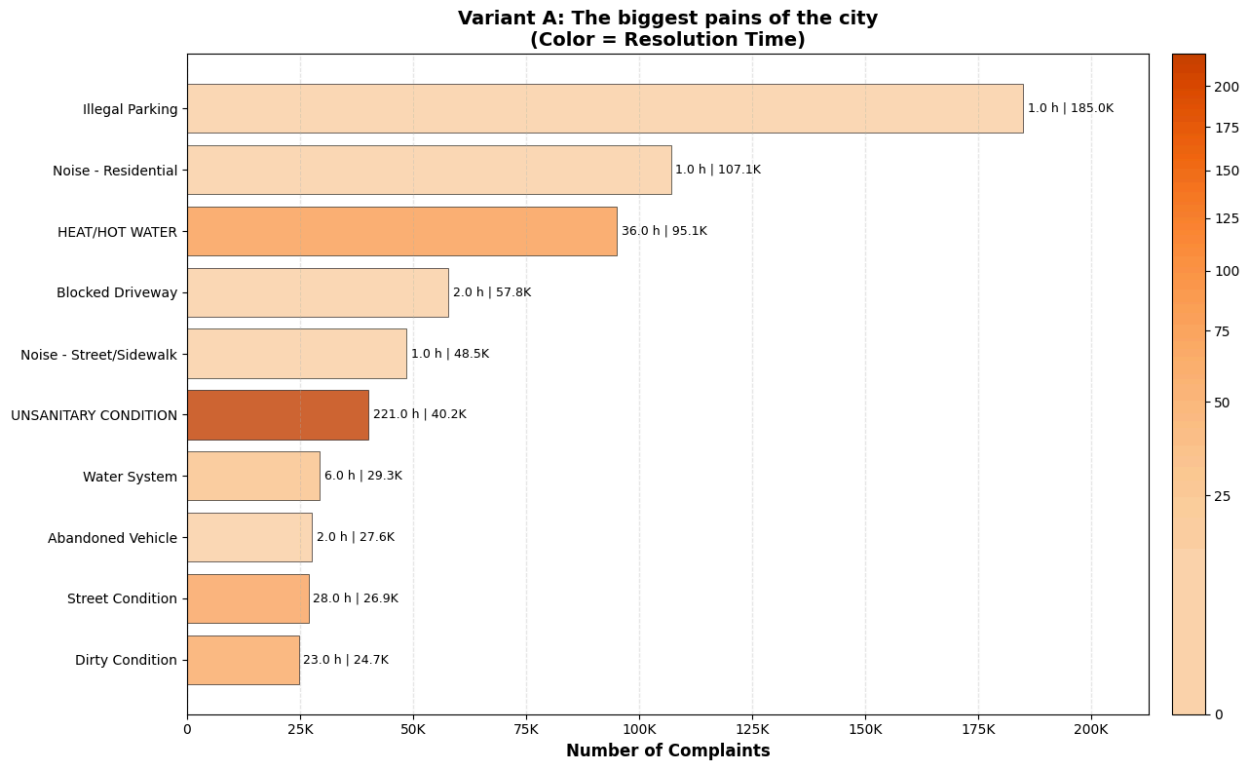


Figure 2.1. Top Delayed Complaint Types




Variant A: The biggest pains of the city
(Color = Resolution Time)
Illegal Parking
Noise - Residential
HEAT/HOT WATER
Blocked Driveway
Noise - Street/Sidewalk
UNSANITARY CONDITION
Water System
Abandoned Vehicle
Street Condition
Dirty Condition
1.0 h | 185.0K
1.0 h | 107.1K
36.0 h | 95.1K
2.0 h | 57.8K
1.0 h | 48.5K
221.0 h | 40.2K
6.0 h | 29.3K
2.0 h | 27.6K
28.0 h | 26.9K
23.0 h | 24.7K
0
25K
50K
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125K
150K
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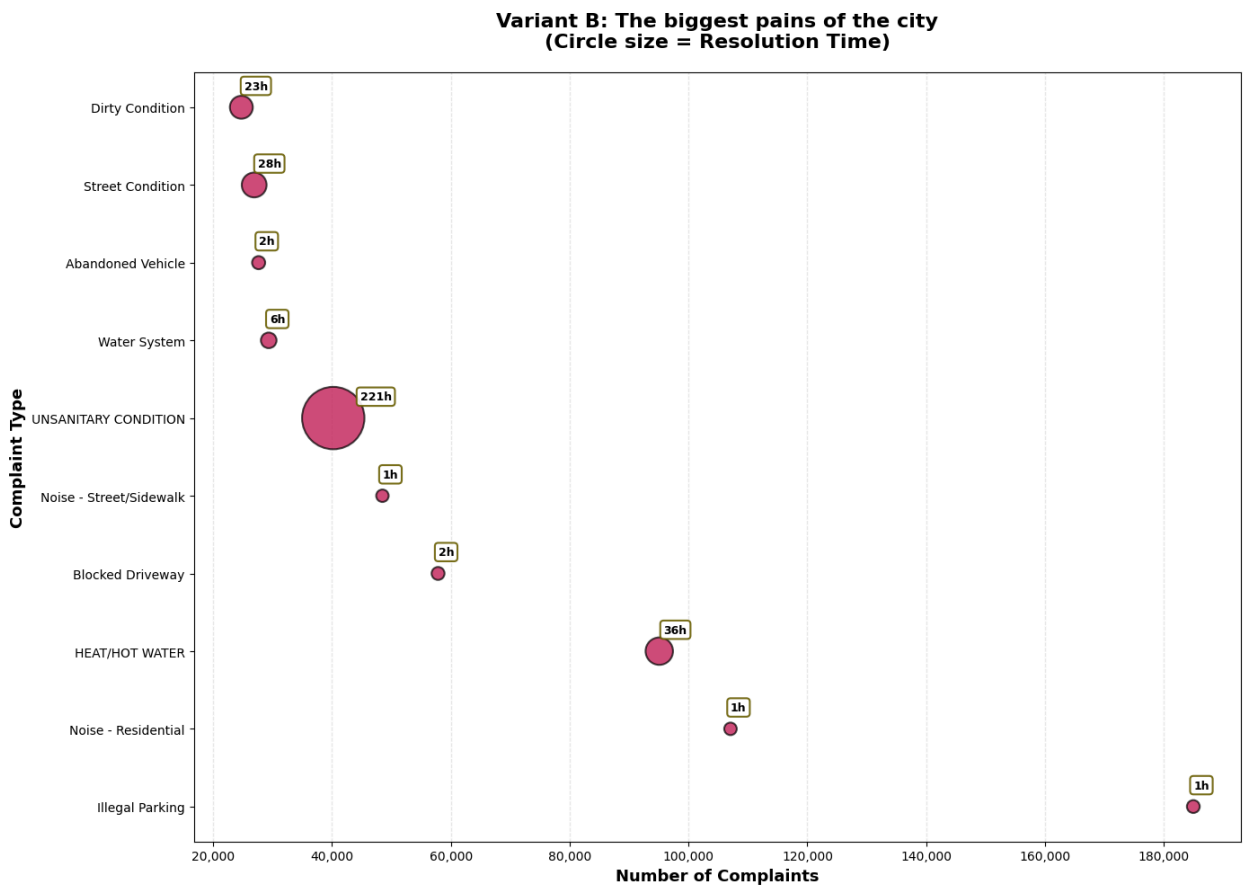
200K
Number of Complaints
0
25
50
75
100
125
150
175
200


Variant B: The biggest pains of the city
(Circle size = Resolution Time)
Complaint Type
Dirty Condition
Street Condition
Abandoned Vehicle
Water System
UNSANITARY CONDITION
Noise - Street/Sidewalk
Blocked Driveway
HEAT/HOT WATER
Noise - Residential
Illegal Parking
23h
28h
2h
6h
221h
1h
2h
36h
1h
1h
20,000
40,000
60,000
80,000
100,000
120,000
140,000
160,000
180,000
Number of Complaints